

Due on monday, December 16th

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1. [4 pts.] Show that an operator, when expressed as a matrix in the basis of its own eigenstates, is diagonal.

Hint: Let's make this more specific: let  $\{|w_n\rangle\}$  be the orthonormalized eigenstates of the operator  $\hat{Y}$ . Use the set  $\{|w_n\rangle\}$  as basis set and express  $\hat{Y}$  as a matrix in this representation. Show that this matrix is diagonal.

2. Consider a system of two spin-1/2 objects. (E.g., the spins of two ions trapped next to each other, or the electron and proton in a hydrogen atom, where we ignore spatial motion and concentrate on their spins.) The Hilbert space is 4-dimensional and is spanned by the basis

$$|\uparrow\uparrow\rangle \quad |\uparrow\downarrow\rangle \quad |\downarrow\uparrow\rangle \quad |\downarrow\downarrow\rangle$$

with the obvious meanings: e.g.,  $|\uparrow\downarrow\rangle$  represents the state where the first spin has  $z$ -component up ( $+\hbar/2$ ) and the second spin has  $z$ -component down ( $-\hbar/2$ ).

Using these four states as the basis set, a general state can be written as

$$|\psi\rangle = \alpha|\uparrow\uparrow\rangle + \beta|\uparrow\downarrow\rangle + \gamma|\downarrow\uparrow\rangle + \delta|\downarrow\downarrow\rangle = \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix}$$

- (a) [2 pts.] In the state  $|\psi\rangle$ , what is the probability of finding the first spin to have  $z$ -component up?
- (b) [2 pts.] What is the probability of finding the second spin to have  $z$ -component down?
- (c) [2 pts.] In a simultaneous measurement of the  $z$ -components of both the spins, one finds the first spin to be spin-up, and the second to be spin-down. What is the state of the system immediately after the measurement?
- (d) [2 pts.] Imagine instead a measurement of only the second spin. The result is found to be spin-up. What is the state of the system immediately after the measurement?

3. (a) [4 pts.] Consider the operator  $\hat{A} = \frac{d}{dx}$ . Which of the following functions are eigenfunctions of  $\hat{A}$ ? (Show your calculation/argument.)

$$f_1(x) = e^{ikx} \quad f_2(x) = ax \quad f_3(x) = \cos(kx) \quad f_4(x) = e^{-ax^2}$$

Here  $k$  and  $a$  are real constants. For those functions that are eigenfunctions, give the eigenvalue.

- (b) [5 pts.] Consider the operator  $\hat{B} = \frac{d^2}{dx^2}$ . Which of the above functions are eigenfunctions of  $\hat{B}$ ? For those functions that are eigenfunctions, give the eigenvalue.

4. The quantum harmonic oscillator.

We denote the orthonormalized energy eigenstates of the harmonic oscillator by  $|n\rangle$ , on which the creation/annihilation operators act as follows:

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle \quad \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

- (a) [7 pts.] Express  $\hat{x}^2$  in terms of  $\hat{a}$  and  $\hat{a}^\dagger$ . Hence show that the uncertainty of the position in the state  $|n\rangle$  is

$$\Delta x = \sigma \sqrt{n + \frac{1}{2}}$$

- (b) [7 pts.] Express  $\hat{p}^2$  in terms of  $\hat{a}$  and  $\hat{a}^\dagger$ . Hence calculate the uncertainty of the momentum,  $\Delta p$ , in the state  $|n\rangle$ .
- (c) [3 pts.] Find the uncertainty product  $\Delta x \Delta p$  in the state  $|n\rangle$ . Also express this quantity as a function of the eigenenergy  $E_n$ .
- (d) [4 pts.] If the system wavefunction at time  $t = 0$  is

$$|\psi(0)\rangle = \frac{\sqrt{3}}{2}|2\rangle - \frac{i}{2}|3\rangle$$

then what is the wavefunction at a later time  $t = T$ ? Your answer should contain the oscillator frequency  $\omega$ .

- (e) [8 pts.] Given that the initial system wavefunction is

$$|\psi(0)\rangle = \frac{\sqrt{3}}{2}|2\rangle - \frac{i}{2}|3\rangle,$$

calculate the expectation value of position in the state  $|\psi(t)\rangle$  as a function of time. Sketch a plot of  $\langle\psi(t)|\hat{x}|\psi(t)\rangle$  against time.

5. (a) [**SELF**] Show that for any three operators  $\hat{A}$ ,  $\hat{B}$ , and  $\hat{C}$ ,

$$[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$$

- (b) [**SELF**] Use the relation  $[\hat{x}, \hat{p}] = i\hbar$  and the result above to prove by mathematical induction the identity

$$[\hat{x}^n, \hat{p}] = in\hbar\hat{x}^{n-1}$$

for any positive integer  $n$ .

(If unfamiliar, please look up “mathematical induction”, a standard technique for proving statements for all positive integers  $n$ .)